

B.Tech in Electrical Engineering

4-year programme · 2024 curriculum onwards

← All UG Programmes
Semester 1
Semester 2
Semester 3
Semester 4
Semester 5
Semester 6
Semester 7
Semester 8
Science Basket
Lab Basket
Optional Independent Projects

 This curriculum applies to students admitted from 2024–25 onwards. If you joined between 2020 and 2023, refer to the [2020–2023 curriculum page](#).

SEMESTER 1 (AUGUST - DECEMBER)			
CODE	COURSE NAME	CREDITS	TYPE
BT1010	Introduction to Life Sciences	1	Basic Sciences
ID1063	Introduction to Programming	3	Basic Engg. Skills
EE1010	Mathematics for Electrical Engineers	3	Basic Sciences
EE1030	Matrix Theory	3	Basic Sciences
EE1101	Circuits and Network Analysis	3	Basic Engg. Skills
LA xxxx	English Communication	2	LA/CA
Total		15	

SEMESTER 2 (JANUARY - MAY)			
CODE	COURSE NAME	CREDITS	TYPE
EE1060	Differential Equations and Transform Techniques	3	Basic Sciences
EE1204	Engineering Electromagnetics	2	Dept. Core
EE1201	Digital Systems	3	Dept. Core
EE1200	Electrical Circuits Lab	2	Lab Basket
EE1080	Probability	3	Basic Engg. Skill
EE1003	Scientific Programming for Electrical Engineers	2	Dept. Core
EE1501	Digital Systems Lab	1	Lab Basket
Total		16	

SEMESTER 3 (AUGUST - DECEMBER)			
CODE	COURSE NAME	CREDITS	TYPE
ID2230	Data Structures & Applications	3	Basic Engg. Skills
EE2000	Signal Processing	3	Dept. Core
EE2103	Semiconductor Device Fundamentals	3	Dept. Core
EE2200	Electrical Machines	3	Dept. Core
EE2080	Random Processes	1	Basic Engg. Skills
(Lab Basket)	Lab Basket	2	Lab Basket
(Sci Basket)	Science Basket	2	Basic Sciences
Total		17	

SEMESTER 4 (JANUARY - MAY)			
CODE	COURSE NAME	CREDITS	TYPE
EE2400	Analog Electronics	3	Dept. Core
EE2600	Power System Analysis	3	Dept. Core
EE2802	Machine Learning	3	Basic Engg. Skills
EE2101	Control Systems	3	Dept. Core
(Sci Basket)	Science Basket	2	Basic Sciences
(Lab Basket)	Lab Basket	2	Lab Basket
EM3020	Introduction to Entrepreneurship	1	LA/CA
Total		17	

SEMESTER 5 (AUGUST - DECEMBER)			
CODE	COURSE NAME	CREDITS	TYPE
EE3301	Intro to VLSI Design	2	Dept. Core
EE3500	Machines Lab	2	Lab Basket
EE3501	Power Electronics	3	Dept. Core
EE3700	Communication Systems	3	Dept. Core
EE3004	Current Trends in EE	1	Dept. Core
(Sci Basket)	Science Basket	2	Basic Sciences
(Free Elec)	Free Elective	1	Free Elective
XX xxxx	LA/CA Elective	2	LA/CA
Total		16	

SEMESTER 6 (JANUARY - MAY)			
CODE	COURSE NAME	CREDITS	TYPE
EE xxxx	Department Electives	12	Dept. Elective
EE xxxx	Lab Basket	2	Lab Basket
EE3014	Current Trends in EE	1	Dept. Core
LA xxxx	LA/CA Elective	2	LA/CA
Total		17	

SEMESTER 7 (AUGUST - DECEMBER)			
CODE	COURSE NAME	CREDITS	TYPE
EE xxxx	Department Electives	6	Dept. Elective
LA xxxx	LA/CA Electives	2	LA/CA
FE xxxx	Free Electives	6	Free Elective
Total		14	

SEMESTER 8 (JANUARY - MAY)			
CODE	COURSE NAME	CREDITS	TYPE
EE xxxx	Department Electives	9	Dept. Elective
LA xxxx	Ethics and Values	1	LA/CA
FE xxxx	Free Electives	6	Free Elective
Total		16	

SCIENCE BASKET (6 CREDITS REQUIRED, SUGGESTED ACROSS SEMESTERS 2-5)			
CODE	COURSE NAME		
EP1108	Modern Physics		
CY1010	Environmental Chemistry		
MA1230	Series of Functions		
MA2130	Complex Variables		
MA2120	Transform Techniques		
MA1130	Vector Calculus		
MA3060	Numerical Analysis		
MA2150	Introduction to Metric Spaces		
EP2200*	Thermodynamics		
EP2108*	Special Relativity		
EP2228*	Fluid Dynamics		
EP2100*	Classical Mechanics		
CC1010*	Introduction to Climate Change		
MA1240*	Combinatorics		
MA2140*	Introduction to Statistics		
MA2510*	Linear Programming		
EP3100	Quantum Mechanics 1		
EE3210	Quantum Mechanics 2		
EP3200	Solid State Physics		
EP3120	Statistical Mechanics		
CY1031	Chemistry Lab		
EP1031	Physics Lab		

LAB BASKET (MINIMUM 6 CREDITS)			
CODE	COURSE NAME		
EE2401	Analog Lab		
EE3401	Microprocessor Lab		
EE3701	Comm Lab		
EE2801	DSP Lab		
EE3802	FPGA Lab		
EE3011	Electromagnetics and Applications Lab		
EE2501	Semiconductor Devices Lab		
EE2301	Electronic Devices and Circuits Lab		
EE2504	Electronic System Design Project Lab		

OPTIONAL INDEPENDENT PROJECTS			
CODE	COURSE NAME		
EE3005	Independent Project		
EE3025	Independent Project		
EE4015	Independent Project		
EE4025	Independent Project		